TC3000J
Technical Communication

Engineering Design and Technical Documents

Dr. Xi Liu

Adjunct Lecturer

Global College, Shanghai Jiao Tong University, Shanghai, China

Engineering Design

Introduction to Engineering Design

"Scientists investigate that which already is; engineers create that which has never been."

-Einstein / Theodore von Kármán?

Introduction to Engineering Design

Mechanical Engineering

Electrical Engineering

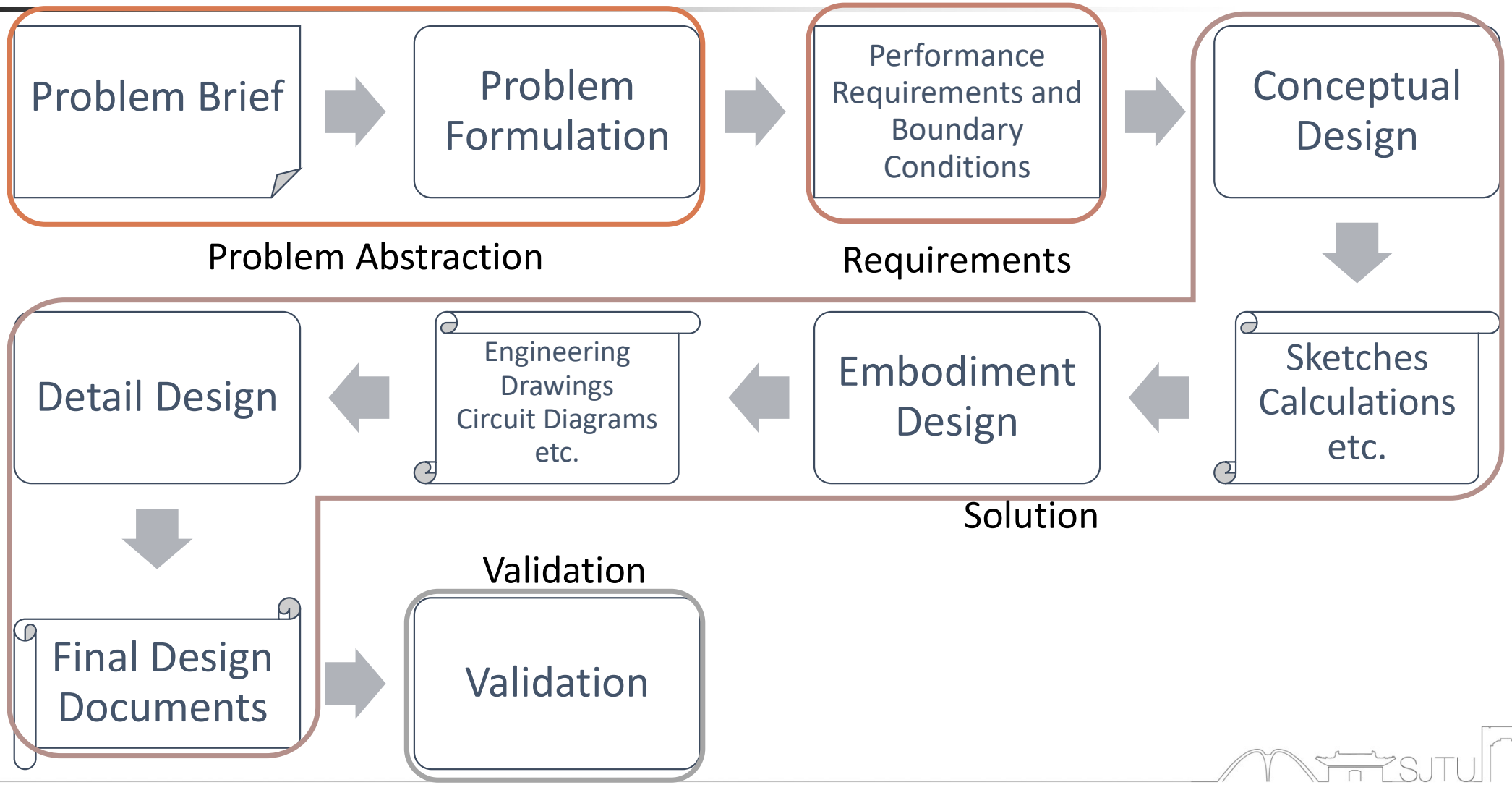
Chemical Engineering

Industrial Engineering

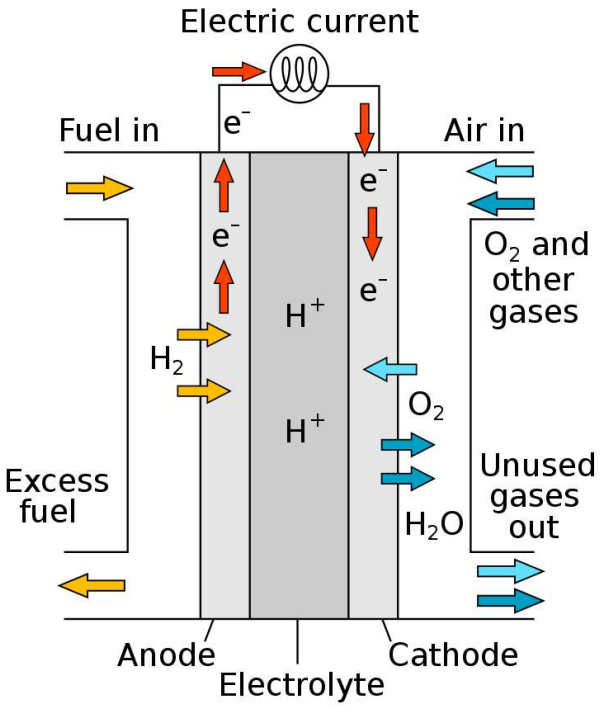
Civil Engineering and Structural Engineering

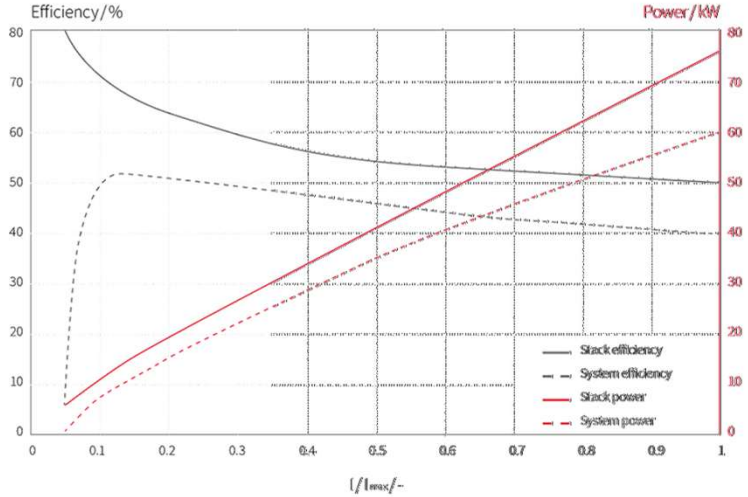
……

Engineering Design Process



Problem-Need-Solution Analysis





Problem-Need-Solution Analysis

Problem	• Negative impact of status quo on people or society
Need	• Specification that directly addresses the problem • Performances

Boundary Conditions	• Variables/conditions that defines the boundaries a design/solution must lie within.
	  • E.g. Minimum or maximum values for a variant/parameter for the design
	  • E.g. Essential conditions that constraints the equation for solution

Solution	• Method by which the need will be met • Specific *how*, addresses the variable in the need
Validation	• Objective assessment of whether the need was met • Checked against the target set by the need

Technical Documentation

Types of Technical Documents

Proposals	Grant applications, activity applications, tenders, patents, cover letters
Technical and Engineering Documents	Design and product specifications, software code, 2D/3D drawings, control diagrams and logic specifications, testing protocols, validation procedures, BOMs
Research and Analysis Documentation	Problem and need analysis report/specifications (SOR, QFD), laboratory and experimental reports, testing protocol, and publications
Instruction Manual	User manuals, service engineer manuals
Visualized Materials for Technical Purposes	Data visualization, Gantt charts, matrices, marketing brochures, project presentations, resumes/CVs
Informational or Boundary Documents	International or national standards (GB, ISO, AMSE, etc.), industrial standards, manufacturing standards, references

Cover Page

Cover pages are important for transmitting documents. They explain what project was completed, for whom, and when.

Laboratory Safety

Editor:	**Xiaoming Wang**
Reviewer:	**Qiang Shi**
Release:	**Qiang Shi**
Version:	**0.1.2**
Status:	**In development**
Date:	**2024-01-28**

M. Liu, "Investigations on the Influence of Oxygen Non-stoichiometry on Stability, Structure and Properties of Hafnium Oxide Thin Films" (2014).
NC State University, Technical Communication Videos, https://www.youtube.com/playlist?list=PLR-UWtI0r_GDI8SOgsyTR4rmW_gtprqB6

Edition Management

Helps to keep track of changes and gives an overview of people responsible for changes.

Edition Management

Version	Date	Editor or Reviewer	Content
0.1.0	2023-11-15	Xiaoming Wang	Document created; parameters defined
0.1.0	2023-11-16	Qiang Shi	Review and correction
0.1.1	2023-12-08	Xiaoming Wang	Addition of analysis section
0.1.2	2024-01-28	Xiaoming Wang	Review and correction according to Da Shi's feedback

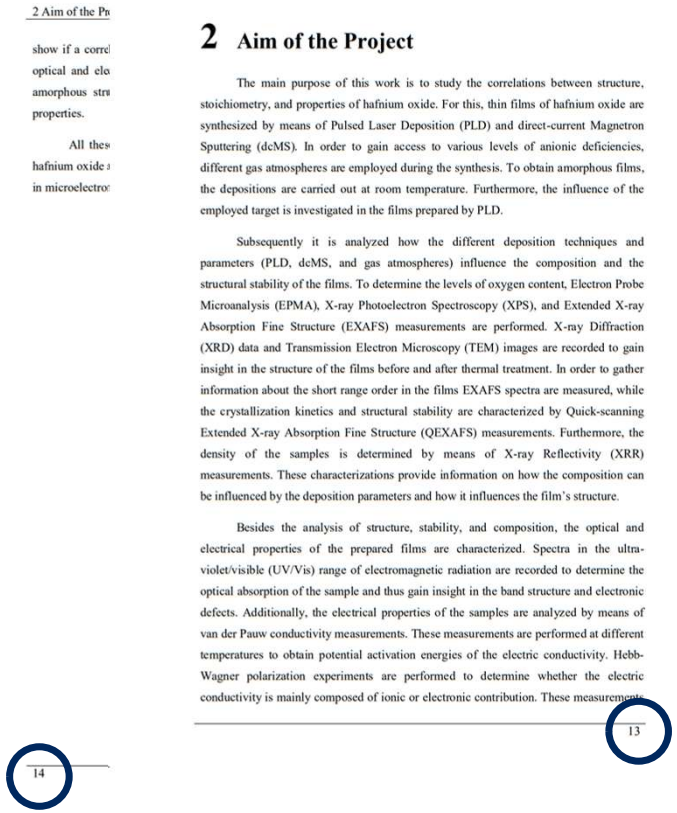
Safety Instructions

- Detailed information on which dangers can arise from the product and which measures need to be taken to avoid these dangers.
- Describes appropriate protective equipment and handling procedures. Usually includes hazard icons (e.g., electrical danger).
- As trivial as heavy weights, as exotic as radioactive or carcinogenic.

https://en.wikipedia.org/wiki/GHS_hazard_pictograms

Page Numbering

- Number pages so your readers can track their progress through your documents.
- You also need them to generate your *Table of Contents*. Use the auto-numbering feature in Microsoft Word, Apple Pages, or Google Docs.



2 Aim of the Pr

show if a corre
optical and ele
amorphous stru
properties.

All these
hafnium oxide a
in microelectro

14

2 Aim of the Project

The main purpose of this work is to study the correlations between structure, stoichiometry, and properties of hafnium oxide. For this, thin films of hafnium oxide are synthesized by means of Pulsed Laser Deposition (PLD) and direct-current Magnetron Sputtering (dcMS). In order to gain access to various levels of anionic deficiencies, different gas atmospheres are employed during the synthesis. To obtain amorphous films, the depositions are carried out at room temperature. Furthermore, the influence of the employed target is investigated in the films prepared by PLD.

Subsequently it is analyzed how the different deposition techniques and parameters (PLD, dcMS, and gas atmospheres) influence the composition and the structural stability of the films. To determine the levels of oxygen content, Electron Probe Microanalysis (EPMA), X-ray Photoelectron Spectroscopy (XPS), and Extended X-ray Absorption Fine Structure (EXAFS) measurements are performed. X-ray Diffraction (XRD) data and Transmission Electron Microscopy (TEM) images are recorded to gain insight in the structure of the films before and after thermal treatment. In order to gather information about the short range order in the films EXAFS spectra are measured, while the crystallization kinetics and structural stability are characterized by Quick-scanning Extended X-ray Absorption Fine Structure (QEXAFS) measurements. Furthermore, the density of the samples is determined by means of X-ray Reflectivity (XRR) measurements. These characterizations provide information on how the composition can be influenced by the deposition parameters and how it influences the film's structure.

Besides the analysis of structure, stability, and composition, the optical and electrical properties of the prepared films are characterized. Spectra in the ultra-violet/visible (UV/Vis) range of electromagnetic radiation are recorded to determine the optical absorption of the sample and thus gain insight in the band structure and electronic defects. Additionally, the electrical properties of the samples are analyzed by means of van der Pauw conductivity measurements. These measurements are performed at different temperatures to obtain potential activation energies of the electric conductivity. Hebb-Wagner polarization experiments are performed to determine whether the electric conductivity is mainly composed of ionic or electronic contribution. These measurements

13

M. Liu, "Investigations on the Influence of Oxygen Non-stoichiometry on Stability, Structure and Properties of Hafnium Oxide Thin Films" (2014).
NC State University, Technical Communication Videos, https://www.youtube.com/playlist?list=PLR-UWtI0r_GDI8SOgsyTR4rmW_gtprqB6

Table of Contents

A Table of Contents (TOC) is essential for along document with multiple sections, headings, and subheading. The TOC helps readers find the information they need quickly and easily.

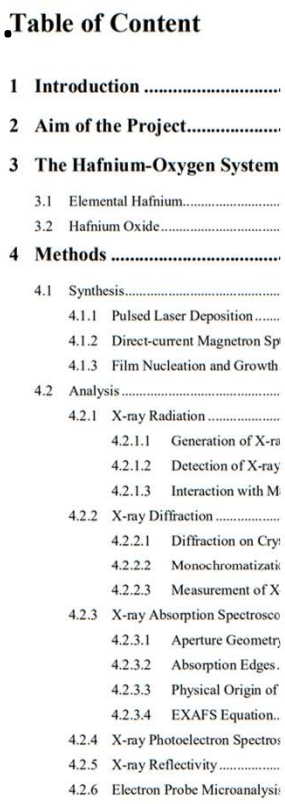
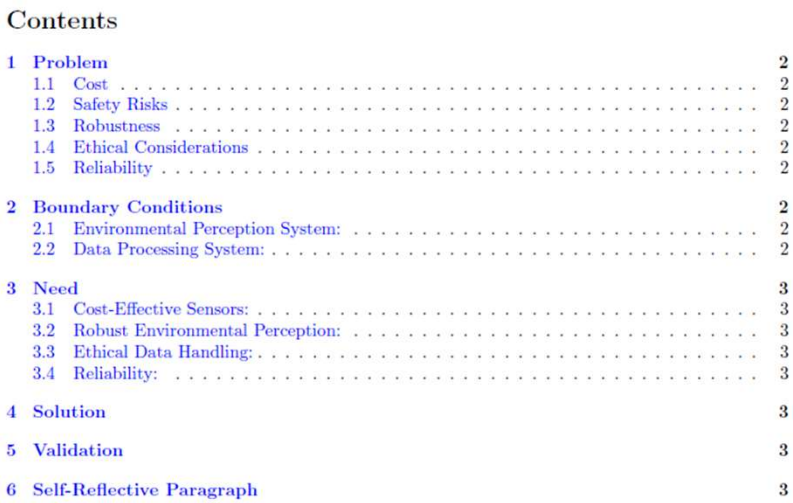


Table of Content

1 Introduction .....
2 Aim of the Project.....
3 The Hafnium-Oxygen System
3.1 Elemental Hafnium.....
3.2 Hafnium Oxide.....
4 Methods .....
4.1 Synthesis.....
4.1.1 Pulsed Laser Deposition.....
4.1.2 Direct-current Magnetron Sp
4.1.3 Film Nucleation and Growth
4.2 Analysis.....
4.2.1 X-ray Radiation .....
4.2.1.1 Generation of X-ra
4.2.1.2 Detection of X-ray
4.2.1.3 Interaction with M
4.2.2 X-ray Diffraction .....
4.2.2.1 Diffraction on Cry
4.2.2.2 Monochromatizati
4.2.2.3 Measurement of X
4.2.3 X-ray Absorption Spectrosco
4.2.3.1 Aperture Geometry
4.2.3.2 Absorption Edges.
4.2.3.3 Physical Origin of
4.2.3.4 EXAFS Equation..
4.2.4 X-ray Photoelectron Spectros
4.2.5 X-ray Reflectivity.....
4.2.6 Electron Probe Microanalysis


Contents

1 Problem 2
1.1 Cost 2
1.2 Safety Risks 2
1.3 Robustness 2
1.4 Ethical Considerations 2
1.5 Reliability 2

2 Boundary Conditions 2
2.1 Environmental Perception System: 2
2.2 Data Processing System: 2

3 Need 3
3.1 Cost-Effective Sensors: 3
3.2 Robust Environmental Perception: 3
3.3 Ethical Data Handling: 3
3.4 Reliability: 3

4 Solution 3

5 Validation 3

6 Self-Reflective Paragraph 3

M. Liu, "Investigations on the Influence of Oxygen Non-stoichiometry on Stability, Structure and Properties of Hafnium Oxide Thin Films" (2014).
NC State University, Technical Communication Videos, https://www.youtube.com/playlist?list=PLR-UWtI0r_GDI8SOgsyTR4rmW_gtprqB6

List of Tables, Figures, etc.

- Lists of Tables, Figures, Equations, Abbreviations, and Symbols give the reader a way to quickly look up certain information in your document.

10 Appendix

10.1 List of Symbols

a_1, a_2, a_3	Cluster shape dependent constants
a, b, c	Lattice parameters of the unit cell [Å]
a_{Avr}	Nucleation rate parameter
A	Cross-sectional area [m^2]
A_{norm}	Normalized absorption
c_{ion}	Ionic charge carrier concentration [$mol \cdot m^{-3}$]
d	Sample thickness [m]
d_l	Lattice layer distance [m]
D	Diffusion coefficient [$m^2 \cdot s^{-1}$]
D_0	Pre-exponential factor [$m^2 \cdot s^{-1}$]
D_{ion}	Diffusion coefficient of ionic charge carriers [$m^2 \cdot s^{-1}$]
E_0	Experimental edge energy [eV]
E_A	Activation energy [eV]
E_F	Fermi energy [eV]
E_M	Mobility edge [eV]
f	Correction factor
F	Faraday constant [96485.3 $C \cdot mol^{-1}$] [127]
$F_s(k)$	Scattering amplitude [Å]
g	Electrode distance [m]

10.2 List of Abbreviations

°	Degree
°C	Degree Celsius [K−273.15]
Å	Ångström [10^{-10} m]
A	Ampere
at%	Atom percent
BD	Blu-ray Disk
BESSY II	Berliner Elektronenspeicherring
CBD	Chemical Bath Deposition
C	Coulomb [A·s]
CD	Compact Disk
CN	Coordination Number
CVD	Chemical Vapor Deposition
d	Day [86,400 s]
dcMS	Direct-current Magnetron Sputtering
DESY	Deutsches Elektronen-Synchrotron
D-MOSTAB	Digital Monochromator Stabilization
DVD	Digital Versatile Disk
DRAM	Dynamic Random Access Memory
EPMA	Electron Probe Microanalysis
eV	Electron Volt [1.60218×10^{-19} J] [127]
EXAFS	Extended X-ray Absorption Fine Structure

10.4 List of Tables

Table 1: Crystal structure parameters of monoclinic hafnium oxide....17
Table 2: Crystal structure parameters of tetragonal hafnium oxide....19
Table 3: Crystal structure parameters of cubic hafnium oxide....20
Table 4: Synthesis parameters for pulsed laser deposition....65
Table 5: Synthesis parameters for direct-current magnetron sputtering....65
Table 6: Film thicknesses and surface roughnesses....88
Table 7: O/Hf ratios of films prepared by different methods....89
Table 8: First shell coordination number fit results....93
Table 9: Resulting parameters from the Avrami fits....105
Table 10: Avrami exponents for different growth modes....106
Table 11: Morphological index and nucleation rate parameter....106
Table 12: Sample densities determined by XRR....107
Table 13: Optical band gaps for samples prepared by different methods....110
Table 14: Activation energies for the electrical conductivity....113

M. Liu, "Investigations on the Influence of Oxygen Non-stoichiometry on Stability, Structure and Properties of Hafnium Oxide Thin Films" (2014).
NC State University, Technical Communication Videos, https://www.youtube.com/playlist?list=PLR-UWtI0r_GDI8SOgsyTR4rmW_gtprqB6

Bibliography

- Also known as List of References. Here all cited and quoted works are listed with full details, so the reader can look up these works and judge their content for themselves.

- Pay attention to the citation format requirements!

9 References

[1] G.D. Wilk, R.M. Wallace, J.M. Anthony, J. Appl. Phys. 89 (2001) 5243.

[2] Y.-H. Tak, K.-B. Kim, H.-G. Park, K.-H. Lee, J.-R. Lee, Thin Solid Films 411 (2002) 12.

[3] A.G. Aberle, Thin Solid Films 517 (2009) 4706.

[4] R.A. Street, Adv. Mater. 21 (2009) 2007.

[5] M. Schulz, Nature 399 (1999) 729.

[6] R. Ghosh, D. Basak, S. Fujihara, J. Appl. Phys. 96 (2004) 2689.

[7] J. Brendt, PhD Thesis, RWTH Aachen University, Aachen, 2011.

[8] A. Elshabini-Riad, F.D. Barlow, Thin Film Technology Handbook, McGraw-Hill, 1998.

[9] M. Wuttig, D. Lusebrink, D. Wamwangi, W. Welnic, M. Gilleszen, R. Dronskowski, Nat. Mater. 6 (2007) 122.

[10] A.V. Kolobov, P. Fons, A.I. Frenkel, A.L. Ankudinov, J. Tominaga, T. Uruga, Nat. Mater. 3 (2004) 703.

[11] K. Nomura, H. Ohta, A. Takagi, T. Kamiya, M. Hirano, H. Hosono, Nature 432 (2004) 488.

[12] A. Takagi, K. Nomura, H. Ohta, H. Yanagi, T. Kamiya, M. Hirano, H. Hosono, Thin Solid Films 486 (2005) 38.

[13] K. Yong Shin, H. Seung-Chul, K. Kim, H. Yang, C. Sung-Yool, Y.T. Kim, J.T. Park, C.H. Lee, J. Choi, J. Paek, K. Lee, Appl. Phys. Lett. 86 (2005) 213105.

[14] M. Ogita, N. Saika, Y. Nakanishi, Y. Hatanaka, Appl. Surf. Sci. 142 (1999) 188.

[15] N. Miura, T. Raisen, G. Lu, N. Yamazoe, Sens. Actuators, B 47 (1998) 84.

[16] V.V. Kharton, A.A. Yaremchenko, A.V. Kovalevsky, A.P. Viskup, E.N. Naumovich, P.F. Kerko, J. Membr. Sci. 163 (1999) 307.

[17] S.P. Harvey, R.A. De Souza, M. Martin, Energy Environ. Sci. 5 (2012) 5803.

[18] D.N. Mueller, R.A. De Souza, T.E. Weirich, D. Roehrens, J. Mayer, M. Martin, Phys. Chem. Chem. Phys. 12 (2010) 10320.

[19] L. Nagarajan, R.A. De Souza, D. Samuelis, I. Valov, A. Borger, J. Janek, K.-D. Becker, P.C. Schmidt, M. Martin, Nat. Mater. 7 (2008) 391.

135

M. Liu, "Investigations on the Influence of Oxygen Non-stoichiometry on Stability, Structure and Properties of Hafnium Oxide Thin Films" (2014).
NC State University, Technical Communication Videos, https://www.youtube.com/playlist?list=PLR-UWtI0r_GDI8SOgsyTR4rmW_gtprqB6

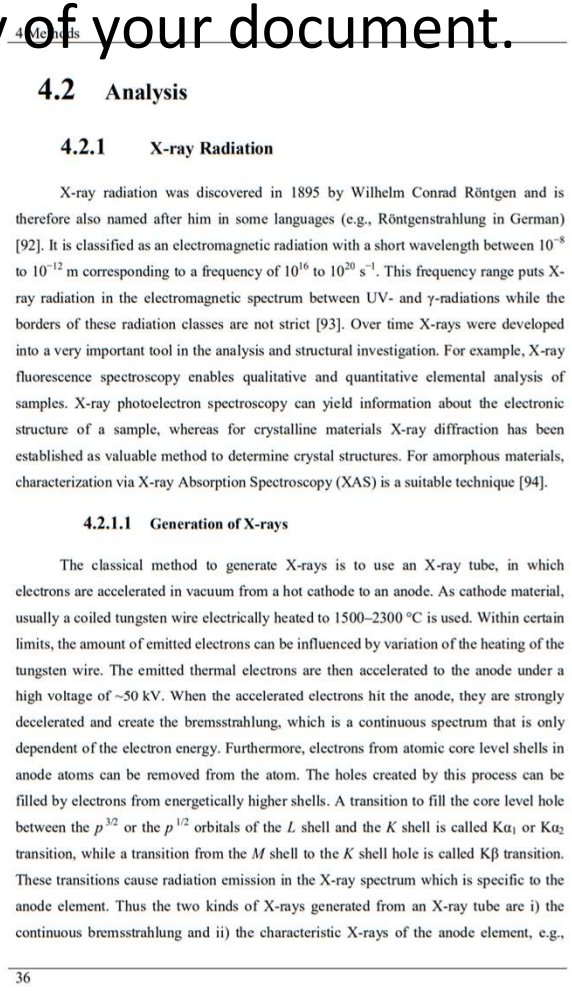
Headings and Text Design

Your text design demonstrates the hierarchy of your document.

Headings

Subheadings

Body text



4 Methods

4.2 Analysis

4.2.1 X-ray Radiation

X-ray radiation was discovered in 1895 by Wilhelm Conrad Röntgen and is therefore also named after him in some languages (e.g., Röntgenstrahlung in German) [92]. It is classified as an electromagnetic radiation with a short wavelength between 10^{-8} to 10^{-12} m corresponding to a frequency of 10^{16} to 10^{20} s^{-1}. This frequency range puts X-ray radiation in the electromagnetic spectrum between UV- and γ-radiations while the borders of these radiation classes are not strict [93]. Over time X-rays were developed into a very important tool in the analysis and structural investigation. For example, X-ray fluorescence spectroscopy enables qualitative and quantitative elemental analysis of samples. X-ray photoelectron spectroscopy can yield information about the electronic structure of a sample, whereas for crystalline materials X-ray diffraction has been established as valuable method to determine crystal structures. For amorphous materials, characterization via X-ray Absorption Spectroscopy (XAS) is a suitable technique [94].

4.2.1.1 Generation of X-rays

The classical method to generate X-rays is to use an X-ray tube, in which electrons are accelerated in vacuum from a hot cathode to an anode. As cathode material, usually a coiled tungsten wire electrically heated to 1500–2300 °C is used. Within certain limits, the amount of emitted electrons can be influenced by variation of the heating of the tungsten wire. The emitted thermal electrons are then accelerated to the anode under a high voltage of ~50 kV. When the accelerated electrons hit the anode, they are strongly decelerated and create the bremsstrahlung, which is a continuous spectrum that is only dependent of the electron energy. Furthermore, electrons from atomic core level shells in anode atoms can be removed from the atom. The holes created by this process can be filled by electrons from energetically higher shells. A transition to fill the core level hole between the $p^{3/2}$ or the $p^{1/2}$ orbitals of the L shell and the K shell is called $K\alpha_1$ or $K\alpha_2$ transition, while a transition from the M shell to the K shell hole is called $K\beta$ transition. These transitions cause radiation emission in the X-ray spectrum which is specific to the anode element. Thus the two kinds of X-rays generated from an X-ray tube are i) the continuous bremsstrahlung and ii) the characteristic X-rays of the anode element, e.g.,

36

M. Liu, "Investigations on the Influence of Oxygen Non-stoichiometry on Stability, Structure and Properties of Hafnium Oxide Thin Films" (2014).
NC State University, Technical Communication Videos, https://www.youtube.com/playlist?list=PLR-UWtI0r_GDI8SOgsyTR4rmW_gtprqB6

Look and Format

Documents should be organized according to audience expectations

Don’t copy prompts from assignment sheets: Use numbered lists or section headings to organize your responses

Use headings and subheadings to help readers understand the information you are presenting

Varies with intention of the document

Use bulleted lists when writing out lists of items

Make sure lists are parallel: Each list item should begin with the same part of speech

Look and Format

Key principle: Documents need a consistent, formal look

Documents should be typed

Use the same font type and size throughout

Recommended: Times New Roman, Arial, Calibri, Aptos, size 12, black

Follow professional formatting standards

For example: 2.55 cm (1”) margin; single spacing; no indented paragraphs; 10 points between paragraphs etc. Specific requirements from journals that is intended for pulishing.